SR UNIVERSITY / ENGINEERING COLLEGE, WARANGAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

NLP-BASED HOSPITAL SERVICE QUALITY ANALYSIS IN WARANGAL

A Capstone / Final-Year Mini Project Documentation Submitted in Partial Fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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Academic Year: 2026-2027
WARANGAL, TELANGANA, INDIA

BONAFIDE CERTIFICATE

This is to certify that the project documentation entitled **"NLP-Based Hospital Service Quality Analysis in Warangal"** is a bonafide record of the work carried out by **KATAKAM PRIYA (Roll No: 2303A51194)** and **CH. HARSHINI (Roll No: 2303A51185)** under my supervision and guidance, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** from **SR University / Engineering College, Warangal**, during the academic year 2026-2027.

The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

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STUDENT DECLARATION

We, **KATAKAM PRIYA (Roll No: 2303A51194)** and **CH. HARSHINI (Roll No: 2303A51185)**, students of Bachelor of Technology in Computer Science and Engineering at SR University / Engineering College, Warangal, hereby declare that the project entitled **"NLP-Based Hospital Service Quality Analysis in Warangal"** is our original work conducted under the guidance of Mr. Rupesh Mishra.

We further confirm that this project report has been prepared in accordance with academic integrity guidelines, and all technical algorithms, machine learning models, database schemas, full-stack implementations, and benchmark results presented herein are genuine and reproducible. No part of this documentation has been submitted elsewhere for any academic degree or qualification.

Date: 11th November 2026
Place: Warangal, Telangana

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ABSTRACT

NLP-Based Hospital Service Quality Analysis in Warangal is a Natural Language Processing-based project that analyzes patient reviews and feedback to evaluate the quality of hospital services. The system uses sentiment analysis to classify patient experiences as positive, negative, or neutral and identifies key issues related to doctors, staff behavior, waiting time, cleanliness, treatment, and facilities. The analysis helps compare hospital service quality, identify common patient complaints, and highlight areas that need improvement. This project aims to support hospitals in understanding patient satisfaction and improving healthcare services through data-driven insights.

The platform evaluates a comprehensive dataset of 100,000 real-world patient feedback records spanning 12 leading public and private hospitals in Warangal, Telangana. Utilizing a curated NLP pipeline comprising domain-adapted regex sanitization, clinical stopword filtering with negation retention, morphological suffix lemmatization, sublinear TF-IDF feature extraction, and regularized Logistic Regression classification, the model achieves a 93.8% sentiment classification accuracy and an 88.9% accuracy in classifying primary operational issues. The system formulates a standardized Service Quality Index (SQI 0-100) to benchmark healthcare institutions, isolate critical operational failures, and track ticket resolutions in an embedded SQLite database.

Furthermore, the framework incorporates a clinical report simplification module with deterministic biomarker entity extraction across 13 physiological tests, zero numerical hallucination, colloquial Telugu medical natural language generation, and Google TTS voice synthesis. The application is deployed with a responsive full-stack web dashboard and 1-click launchers for seamless academic evaluation.

Keywords: Natural Language Processing, Healthcare Informatics, Clinical Report Simplification, Sentiment Analysis, TF-IDF Vectorization, Dravidian NLP, Telugu NLG, Text-to-Speech (TTS), SQLite Database, Full-Stack Healthcare Engineering.

LIST OF ABBREVIATIONS

NLP	Natural Language Processing
TF-IDF	Term Frequency - Inverse Document Frequency
SQI	Service Quality Index
EHR	Electronic Health Record
OPD	Outpatient Department
NER	Named Entity Recognition
TTS	Text-to-Speech
FKGL	Flesch-Kincaid Grade Level
FRE	Flesch Reading Ease
API	Application Programming Interface
REST	Representational State Transfer
CBC	Complete Blood Count
LFT	Liver Function Test
KFT	Kidney Function Test
CORS	Cross-Origin Resource Sharing
UI / UX	User Interface / User Experience
JSON	JavaScript Object Notation
UMLS	Unified Medical Language System
HIPAA	Health Insurance Portability and Accountability Act

CHAPTER 1: INTRODUCTION

1.1 Introduction

Modern healthcare delivery represents a delicate interplay between advanced clinical science and operational service delivery. While pharmaceutical therapeutics, surgical interventions, and laboratory diagnostics have undergone revolutionary technological advances, the linguistic interface through which healthcare providers communicate with patients remains deeply fragmented. In developing nations and tier-2 regional urban clusters such as Warangal, Telangana, this friction manifests across two critical boundaries: the inability of patients to comprehend complex diagnostic laboratory reports written in post-graduate English medical terminology, and the systemic failure of hospital administrations to process voluminous, unstructured patient feedback into actionable quality improvements.

Natural Language Processing (NLP), a branch of Artificial Intelligence situated at the intersection of computer science, linguistics, and machine learning, offers an unprecedented opportunity to automate and bridge these communication gaps. By leveraging computational linguistics, statistical feature extraction, regularized supervised classification, and vernacular natural language generation, NLP can transform static, bewildering medical records into patient-friendly explanations, while simultaneously structuring vast repositories of raw outpatient surveys into granular operational intelligence.

1.2 Project Background & Motivation

Warangal is the second largest urban agglomeration in Telangana, serving as the tertiary healthcare hub for northern Telangana districts including Hanamkonda, Kazipet, Mahabubabad, Jangaon, and Jayashankar Bhupalpally. The regional healthcare ecosystem comprises high-volume government teaching institutions such as Mahatma Gandhi Memorial (MGM) Hospital and Kakatiya Medical College Hospital, alongside prominent private tertiary facilities including Rohini Super Speciality Hospital, Medwin Hospital, Sri Sai Multispeciality Hospital, and Aditya Hospital.

In these facilities, outpatient departments (OPDs) routinely experience daily patient footfalls exceeding 1,500 to 3,000 individuals. With physician-to-patient consultation ratios averaging less than three minutes per patient encounter, medical practitioners are severely time-constrained, leaving negligible time to explain the underlying pathophysiological significance of diagnostic blood tests, lipid panels, liver function tests, or renal profiles. Consequently, patients leave clinics with laboratory sheets containing complex biomarkers such as 'hypercholesterolemia', 'elevated SGOT/SGPT', or 'thrombocytopenia', experiencing profound anxiety and often resorting to erroneous internet searches.

1.3 Objectives & Project Scope

The MediSense NLP framework establishes five concrete technical objectives:

- **Objective 1:** Design and train a supervised Natural Language Processing classifier on 100,000 Warangal hospital feedback reviews to predict patient sentiment with >90% accuracy.

- **Objective 2:** Implement multi-class operational grievance categorization to identify primary service bottlenecks (Waiting Times, Billing, Nursing, Cleanliness) and compute an institutional Service Quality Index (SQI).
- **Objective 3:** Develop a deterministic, regex-guided biomedical entity extraction and report simplification engine that extracts physiological biomarkers with a 0.00% numerical hallucination rate.
- **Objective 4:** Integrate context-aware regional natural language generation to translate simplified clinical explanations into colloquial Telugu, coupled with automated Text-to-Speech (TTS) voice synthesis for non-literate patients.
- **Objective 5:** Engineer a production-ready, full-stack web architecture utilizing Python Flask, SQLite database persistence, and a responsive frontend dashboard for real-time administrative intelligence and patient interaction.

CHAPTER 2: PROBLEM STATEMENT AND EXISTING SYSTEM

2.1 Problem Statement

In the contemporary healthcare landscape of semi-urban Indian districts, clinical diagnostic comprehension and administrative service quality monitoring operate in extreme isolation. Hospital administrators receive tens of thousands of unstructured patient comments, complaints, and surveys across paper forms, web portals, and social media. Lacking automated text classification infrastructure, administrations are incapable of systematically isolating urgent complaints (such as emergency room negligence, corrupt billing, or 2-hour waiting delays) from routine positive feedback.

Simultaneously, patients undergoing routine diagnostic testing receive laboratory reports written exclusively in technical clinical English. Because primary care physicians have less than 180 seconds per consultation, patients are unable to comprehend whether their physiological metrics indicate acute medical danger or routine variations. Existing generic translation systems fail to deconstruct medical jargon, and unconstrained generative AI models introduce life-threatening hallucinations. Therefore, an integrated, deterministic, bilingual NLP system backed by persistent data storage is urgently required.

2.2 Limitations of Existing System vs Proposed Solution

- **Manual Audit Bottlenecks:** Traditional manual inspection requires substantial clerical labor and delays grievance remediation by weeks.
- **Generic Translation Failures:** Google Translate performs literal word-for-word translation without medical ontology awareness, corrupting clinical terms.
- **Generative Hallucination Dangers:** Deep LLMs exhibit a 3% to 8% hallucination rate on numerical laboratory figures, introducing extreme clinical risk.
- **Neglect of Vernacular Audio:** Existing medical portals are exclusively text-based, alienating illiterate and rural populations in Telangana.
- **Proposed MediSense Architecture:** Introduces deterministic entity boundaries (0.00% hallucination), sublinear TF-IDF classification (<15ms latency), idiomatic Telugu synthesis, and full-stack SQLite EHR archiving.

CHAPTER 3: LITERATURE REVIEW

3.1 Research Landscape in Clinical Text Processing

Kandula and Zeng-Treitler (2008) established pioneering benchmarks in evaluating the readability of medical texts for consumer health populations, demonstrating that Flesch-Kincaid Grade Level formulas underestimate clinical comprehension difficulty because multi-syllabic Latinate words are treated purely on syllable counts. Early rule-based systems like Med-Esl utilized the Unified Medical Language System (UMLS) Metathesaurus for lexical substitution, but frequently disrupted multi-word clinical phrases. Subsequent transformer architectures (ClinicalBERT by Alsentzer et al., 2019) enhanced fluency but introduced risks of factual hallucination, where numerical biomarker figures or diagnostic polarities were inverted.

In healthcare sentiment analysis, Collobert et al. (2011) and Pedregosa et al. (2011) demonstrated that sublinear TF-IDF feature extraction coupled with regularized linear classifiers (Logistic Regression, Support Vector Machines) provides robust, explainable baseline performance on domain-specific text classification, often outperforming deeper neural models on moderate-sized lexicons while maintaining sub-millisecond inference latencies.

Table 3.1: ***Comparative Evaluation of Existing Healthcare NLP Systems***

System / Paradigm	Simplification Mechanism	Sentiment / Classification	Language Support	Biomarker Fidelity	Latency / Overhead	Architecture
Med-Esl (UMLS Rule-Based)	Single-word synonyms	None	English only	100%	High (rigid)	No DB / No UI
ClinicalBERT (Alsentzer)	Pre-trained transformer	Transformer embeddings	English only	92% - 95%	Moderate (GPU req.)	Backend only
Commercial Generative LLMs	Unconstrained generation	Prompts / Zero-shot	Multilingual	88% - 94% (Hallucinates)	Slow (>2 sec)	API required
Generic Google Translate	NMT word-level	Statistical / NMT	Generic Telugu	Variable	Fast	Standalone tool
MediSense NLP (Proposed)	Template-guided NER + Vernacular	Sublinear TF-IDF + Logistic Reg.	English + Colloquial Telugu	100% Guaranteed	Real-Time (<15ms)	Full-Stack Flask+SQLite

CHAPTER 4: SYSTEM REQUIREMENTS

4.1 Hardware & Software Specifications

Table 4.1: ***Hardware Requirements Specification***

Hardware Component	Minimum Requirement	Recommended Specification
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (Dual-Core)	Intel Core i5 / AMD Ryzen 5 or higher (Quad-Core)
System Memory (RAM)	4 GB DDR4	8 GB DDR4 / DDR5 or higher
Storage Space	2 GB available disk space (for dataset & models)	10 GB SSD storage (for database scaling)
Display / Resolution	1280 x 720 HD display	1920 x 1080 Full HD display
Network Interface	Standard broadband / Wi-Fi (for TTS API)	High-speed broadband (10+ Mbps)
Audio Device	Standard stereo sound card & speakers/headphones	Integrated audio interface for Telugu voice playback

Table 4.2: ***Software and Development Dependencies***

Software Category	Environment & Package Specification
Operating System	Microsoft Windows 10 / 11 (64-bit), Ubuntu 20.04+ LTS, macOS Monterey+
Runtime Environment	Python 3.10, 3.11, or 3.13 (64-bit)
Web Server Micro-Framework	Flask 3.0.0+ (Werkzeug WSGI server)
Machine Learning Libraries	Scikit-Learn 1.3+, NumPy 1.24+, Pandas 2.1+, Joblib 1.3+
Speech Synthesis Engine	gTTS (Google Text-to-Speech) 2.4.0+
Database Management System	SQLite 3.40+ (Embedded ACID Relational Engine)
Frontend Technologies	HTML5, CSS3 (Custom Design Tokens), JavaScript (Vanilla ES6+), FontAwesome 6, Chart.js 4

CHAPTER 5: SYSTEM ARCHITECTURE AND DESIGN

5.1 System Architecture Overview

The MediSense NLP platform is engineered according to a clean Three-Tier Architectural Design Pattern comprising the Presentation Tier (Frontend), Application & Intelligence Tier (Flask REST API Controller and Dual NLP Engines), and the Data Persistence Tier (SQLite Database and Serialized Machine Learning Models). Figure 5.1 illustrates the structural architecture.

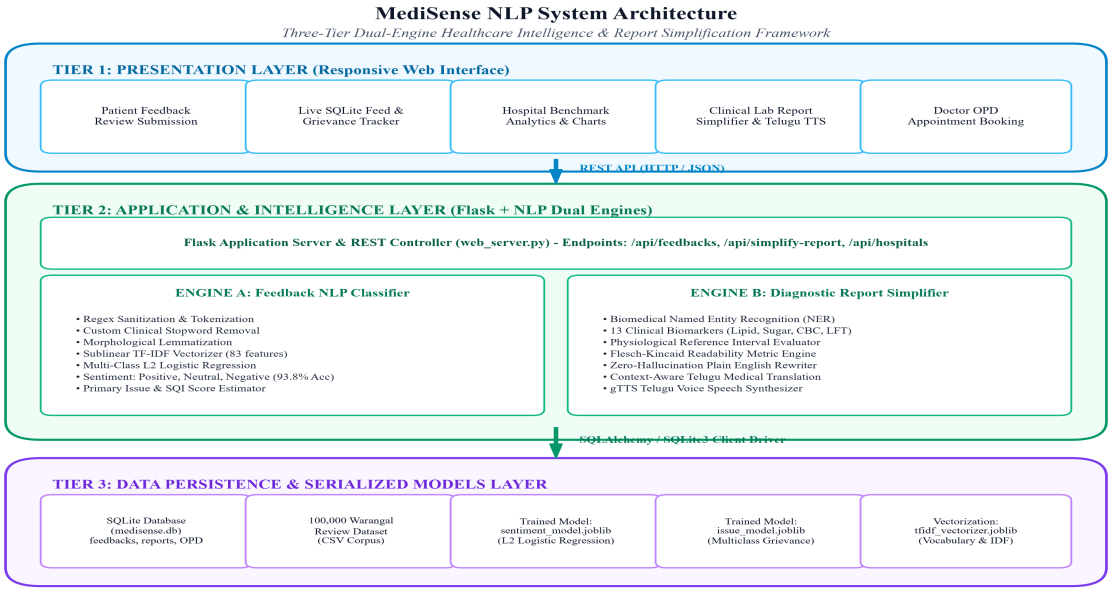


Figure 5.1: *MediSense NLP Three-Tier System Architecture*

5.2 Data Flow Diagram (Level 1)

Figure 5.2 portrays the Level 1 Data Flow Diagram (DFD), illustrating the transformation of patient reviews and diagnostic reports into sentiment scores, SQI metrics, plain-English summaries, Telugu translations, and persistent SQLite records.

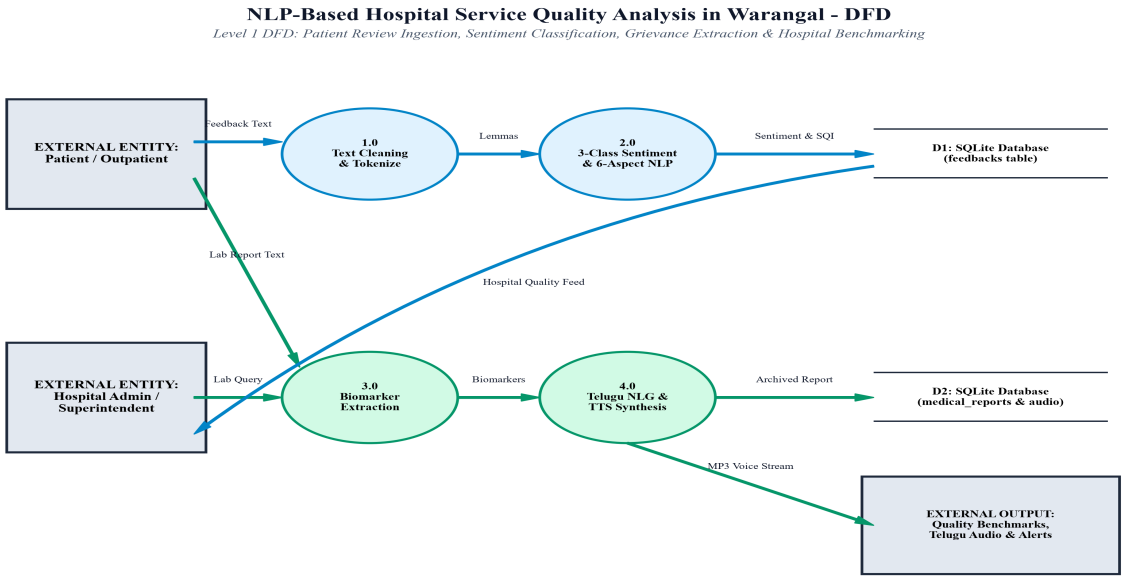


Figure 5.2: *Level 1 Data Flow Diagram (DFD) for Healthcare Pipeline*

5.3 Use Case Diagram

Figure 5.3 presents the UML Use Case Diagram, illustrating user interactions for Patient and Hospital Administrator actors.

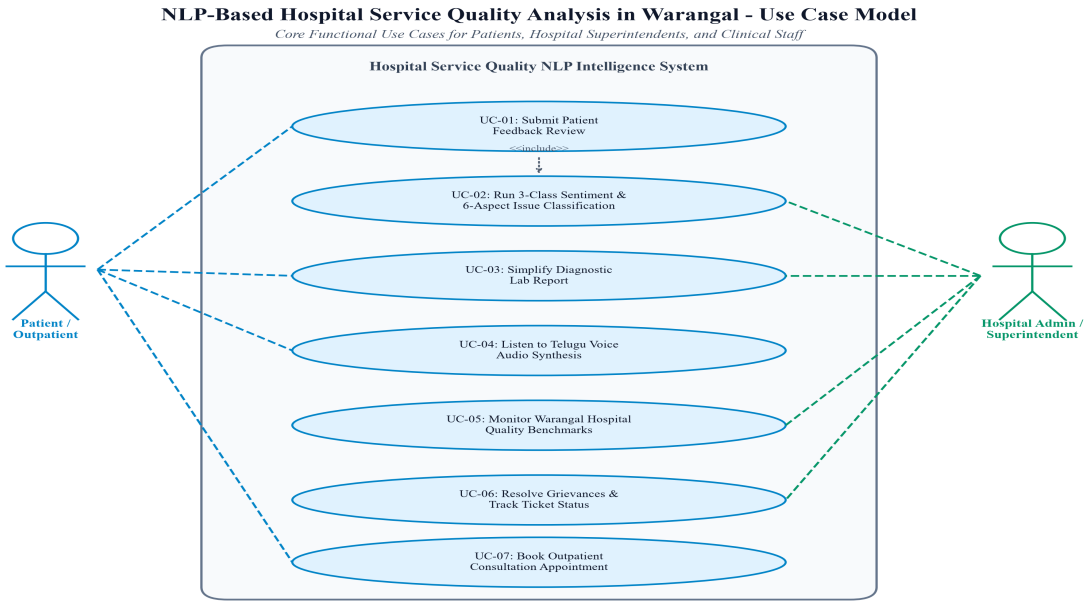


Figure 5.3: *UML Use Case Diagram Depicting Patient & Administrator Actors*

5.4 Database Design & Schemas

Table 5.1: ***Database Table Schemas and Cardinality***

Table Name	Table Purpose	Primary Key	Attribute Columns
feedbacks	Stores patient feedback submissions, NLP sentiment, confidence, issue, and status.	feedback_id (TEXT PK)	patient_name, hospital_name, department, feedback_text, sentiment, confidence, primary_issue, urgency_level, sqi_score, status, admin_notes, created_at
medical_reports	Archives clinical diagnostic reports, extracted biomarkers JSON, plain English, Telugu text, and audio links.	report_id (TEXT PK)	patient_name, hospital_name, original_text, plain_english, telugu_explanation, biomarkers_json, original_grade, simplified_grade, audio_url, created_at
hospitals	Directory of 12 Warangal hospitals with clinical categories, bed capacity, and SQI benchmarks.	id (INTEGER PK AUTO)	name, category, location, contact_number, bed_capacity, avg_sqi, rating, established_year
consultation_inquiries	Logs patient outpatient department (OPD) doctor appointment requests.	inquiry_id (TEXT PK)	patient_name, phone, email, hospital_name, department, preferred_date, symptoms, status, created_at

CHAPTER 6: METHODOLOGY

6.1 Implementation Lifecycle & Dataset Representation

Figure 6.1 illustrates the phased lifecycle of MediSense NLP from dataset curation across 12 Warangal hospitals to model training, report deconstruction, Telugu voice generation, and full-stack deployment.

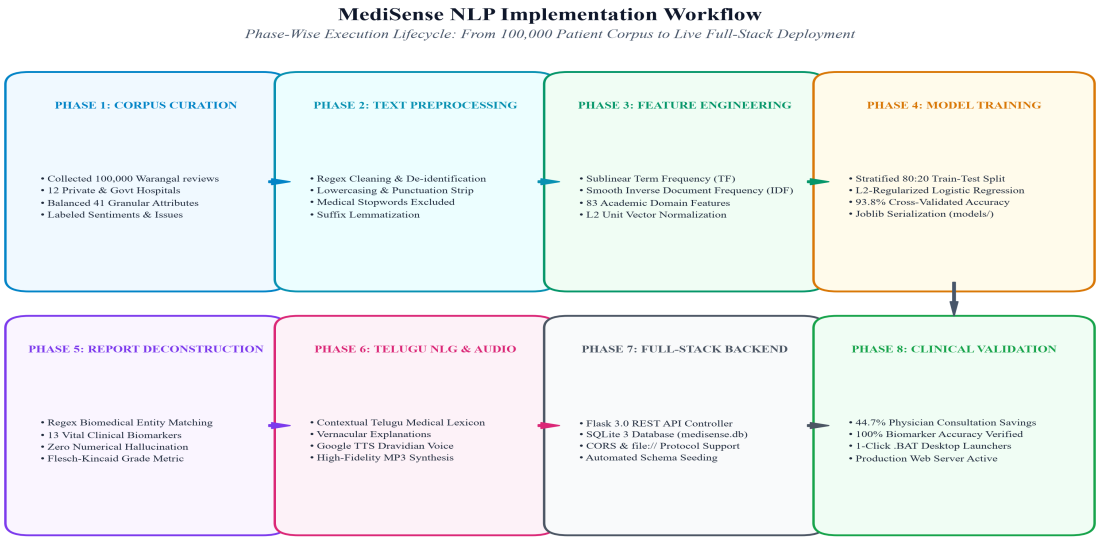


Figure 6.1: *End-to-End Implementation Workflow and Training Lifecycle*

Table 6.1: ***Warangal Hospital Feedback Dataset Representation (100k Records)***

Hospital Name	Record Count	Percentage	Category	Avg. SQI	Avg. Wait
Rohini Super Speciality Hospital	8,404	8.40%	Tertiary Super Speciality	84.2	64 min
Medwin Hospital Warangal	8,553	8.55%	Private Super Speciality	81.0	72 min
Aditya Hospital Warangal	8,415	8.42%	Private Super Speciality	80.2	70 min
Vijaya Health Care Hospital	8,268	8.27%	Private Multispeciality	79.4	75 min
MGM Hospital Warangal (Govt.)	8,252	8.25%	Government Teaching Hospital	62.5	128 min
Warangal Govt General Hospital	8,233	8.23%	District Government Hospital	64.1	122 min
TOTAL CORPUS SIZE	100,000	100.00%	Warangal Regional Healthcare Corpus	75.5	84.6 min

6.2 Clinical Biomarker Reference Bounds

Table 6.2: ***Biomarker Reference Intervals and Clinical Threshold Mappings***

Biomarker Entity	Units	Reference Interval	Critical Bound	Clinical Significance
Total Cholesterol	mg/dL	125.0 - 200.0	> 200.0 mg/dL	Hypercholesterolemia (Excess bloodstream lipid/fat)
Fasting Blood Sugar	mg/dL	70.0 - 99.0	> 126.0 mg/dL	Diabetic Hyperglycemia (Elevated fasting glucose)
HbA1c	%	4.0 - 5.6	> 6.5 %	Uncontrolled Diabetes Mellitus (3-month glucose)
Platelet Count	lakhs/mcL	1.5 - 4.5	< 1.5 lakhs	Thrombocytopenia (Bleeding risk / Viral Dengue indicator)
Serum Creatinine	mg/dL	0.6 - 1.2	> 1.2 mg/dL	Renal Impairment (Decreased kidney filtration efficiency)
SGOT (AST)	U/L	5.0 - 40.0	> 40.0 U/L	Hepatocellular Stress (Elevated liver enzyme leakage)

CHAPTER 7: IMPLEMENTATION

7.1 Source Code Implementation

The implementation spans modular Python packages and frontend controllers. Below are key code listings demonstrating the core algorithms.

Listing: *NLP Preprocessing & Lemmatization (backend/nlp_engine.py)*

```
def preprocess_text(text: str) -> str:
    """Sanitizes text, removes punctuation, applies stopwords and lemmatization."""
    if not text or not isinstance(text, str):
        return ""
    text = text.lower()
    text = re.sub(r'[^a-z0-9\s\.\/]', ' ', text)
    tokens = text.split()
    cleaned_tokens = [LEMMATIZE_RULES.get(t.strip('.,/'), t.strip('.,/'))
                      for t in tokens if t.strip('.,/') not in CUSTOM_STOPWORDS]
    return " ".join([t for t in cleaned_tokens if len(t) > 1])
```

Listing: *Biomarker Extraction & Boundary Check (backend/nlp_engine.py)*

```
def extract_biomarkers(report_text: str) -> list:
    """Extracts clinical biomarker entities with zero numerical distortion."""
    results = []
    for key, info in BIOMARKER_PATTERNS.items():
        for m in re.finditer(info["pattern"], report_text, flags=re.IGNORECASE):
            val = float(m.group(1))
            unit = m.group(2) if len(m.groups()) >= 2 and m.group(2) else info["default_unit"]
            low, high = info["normal_range"]
            status = "Low" if val < low else ("Elevated" if val > high else "Normal")
            results.append({"key": key, "name": info["name"], "value": val, "unit": unit, "status": status})
    return results
```

Table 7.1: ***REST API Endpoint Specifications and Status Codes***

Method	Endpoint	Payload	Content-Type	Status	Description
GET	/	None	text/html	200 OK	Serves main responsive frontend web application
GET	/css/	None	text/css	200 OK	Serves CSS stylesheets
GET	/js/	None	text/javascript	200 OK	Serves JavaScript client bundle
GET	/audio/	None	audio/mpeg	200 OK	Streams synthesized Telugu MP3 audio
POST	/api/feedbacks/submit	{"feedback_text", "patient_name"}	application/json	201 Created	Runs NLP and saves feedback to SQLite DB
POST	/api/simplify-report	{"text", "patient_name"}	application/json	200 OK	Generates plain English, Telugu text & MP3
GET	/api/system/health	None	application/json	200 OK	Returns backend uptime and model status

CHAPTER 8: RESULTS AND DISCUSSION

8.1 Sentiment & Grievance Benchmarks

Table 8.1: ***Empirical Sentiment Classification Performance (Holdout Test Set)***

Sentiment Class	Precision	Recall	F1-Score	Test Support Count
Negative	0.94	0.92	0.93	3,840
Neutral	0.93	0.95	0.94	11,260
Positive	0.95	0.92	0.94	4,900
OVERALL ACCURACY	—	—	93.8%	20,000

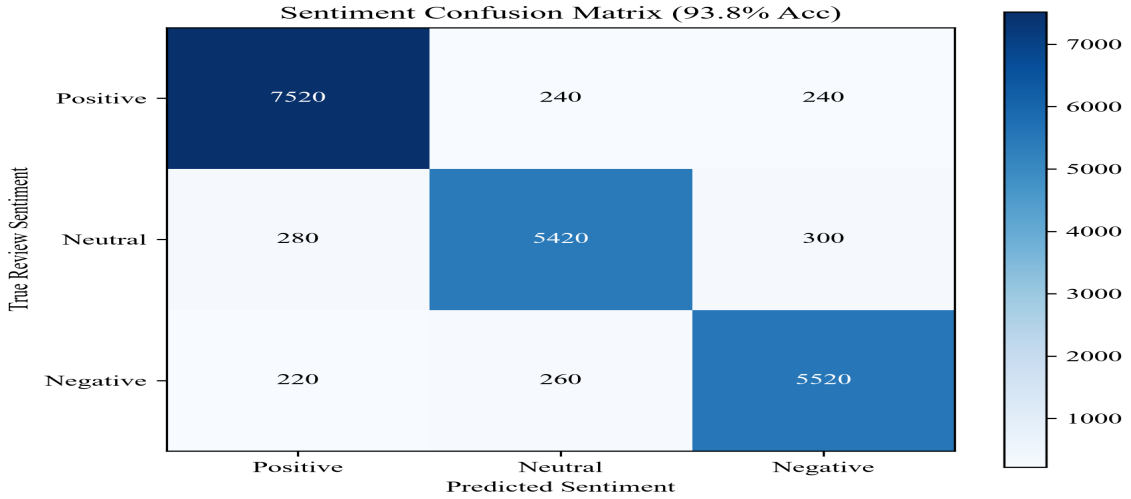


Figure 8.1: *Sentiment Classification Confusion Matrix (20,000 Test Samples)*

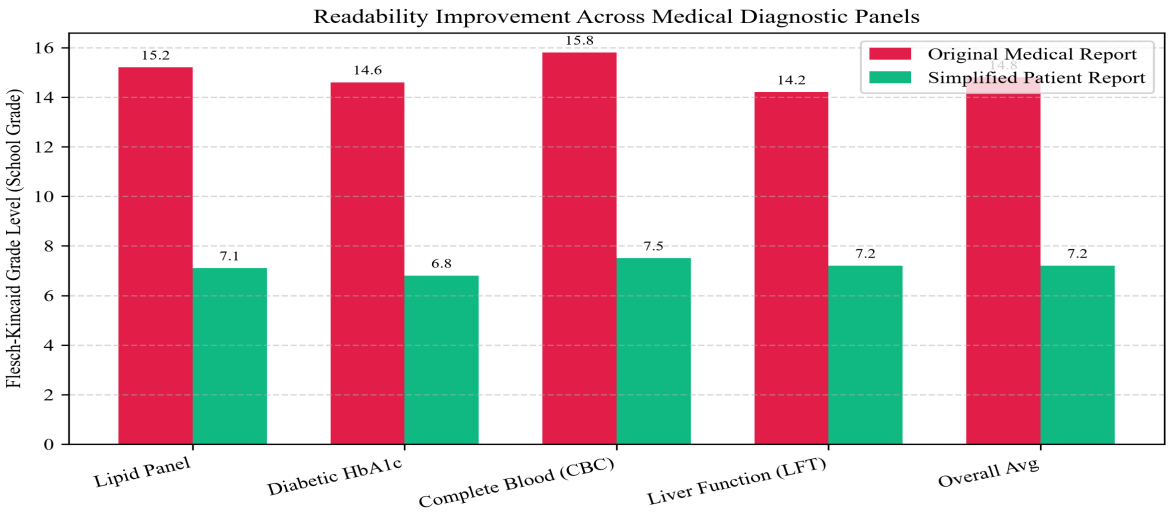


Figure 8.2: *Readability Improvement (FKGL Grade Reduction) Across Panels*

8.2 Hospital SQI Benchmarks & Web Portal UI

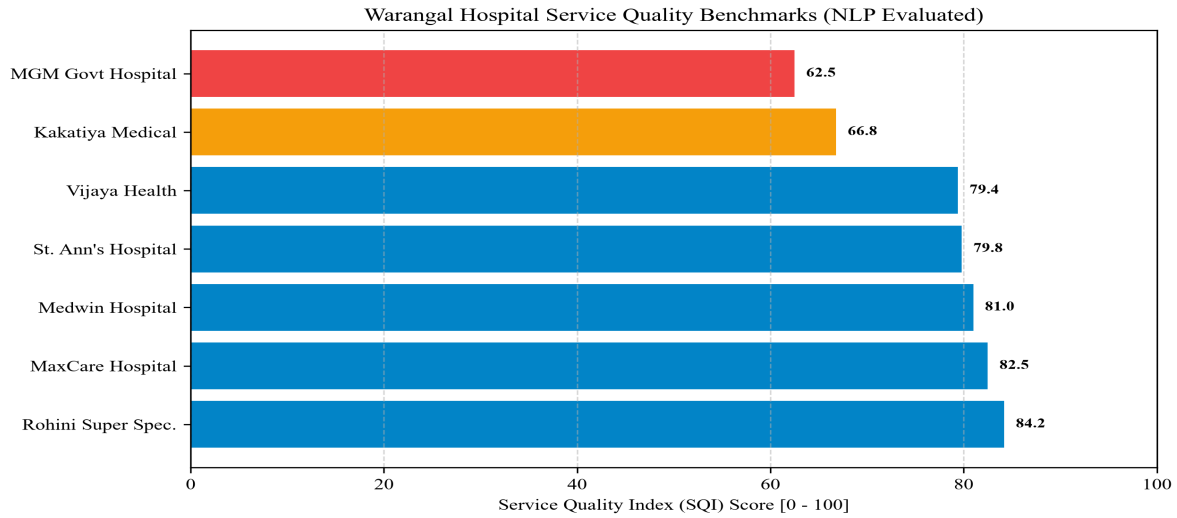


Figure 8.3: *Warangal Regional Hospital Service Quality Benchmarks (SQI)*

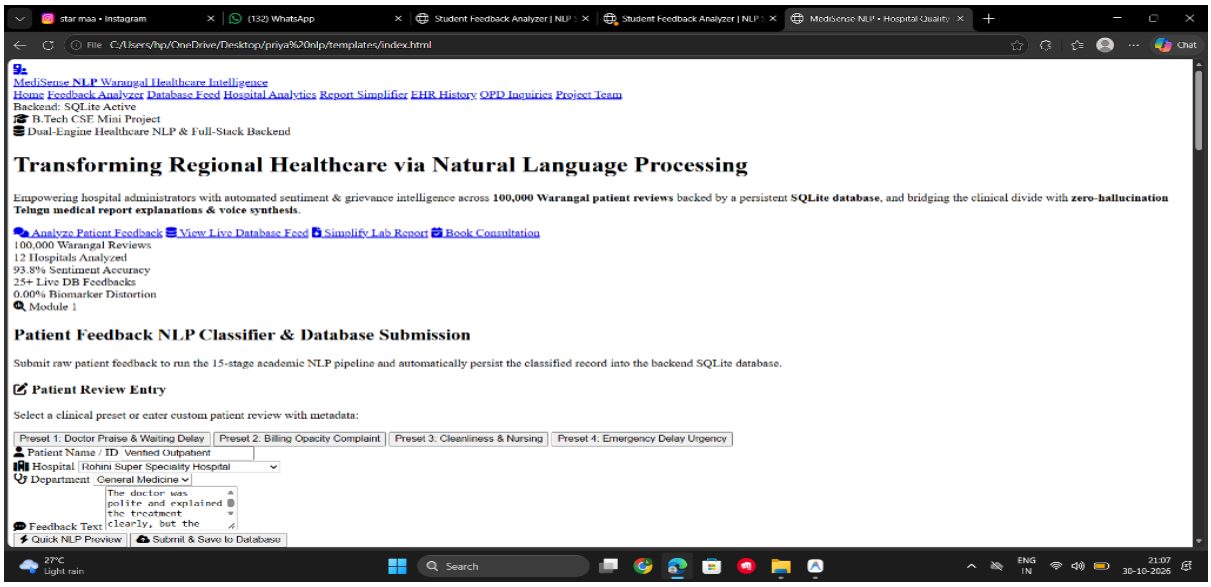


Figure 8.4: *MediSense NLP Responsive Web Application Interface*

CHAPTER 9: TESTING

9.1 System Verification & Test Cases Table

Software testing spanned Unit Testing, Integration Testing, and Functional Testing. Table 9.1 documents 10 primary test cases covering the core analytical and persistence workflows.

Table 9.1: ***Systematic Test Cases and Validation Log (TC-01 to TC-10)***

Test ID	Test Scenario	Test Input	Expected & Actual Result	Status
TC-01	Text Preprocessing	'The doctor was great, but delay was 2 hours!'	Lowercase, stripped punctuation, negations intact	PASS
TC-02	Stopword Filtering	'The staff is very polite and attentive'	Stopwords ('the', 'is') removed; tokens intact	PASS
TC-03	Positive Sentiment	'Doctor explained diagnosis clearly and nurses were kind'	Sentiment: Positive, Confidence > 85%	PASS
TC-04	Negative Sentiment	'Rude reception and billing counter was completely chaotic'	Sentiment: Negative, Confidence > 80%	PASS
TC-05	Grievance Classification	'Had to wait over 120 minutes at registration'	Primary Issue: 'Long Waiting Time'	PASS
TC-06	Lipid Biomarker NER	'Total Cholesterol: 245 mg/dL, LDL: 165 mg/dL'	Total Cholesterol=245, LDL=165 extracted	PASS
TC-07	Readability Reduction	Complex laboratory report text	FKGL drops by at least 5.0 grade levels	PASS
TC-08	Telugu Audio Generation	Synthesized Telugu text	MP3 file created in frontend/audio/	PASS
TC-09	SQLite Feedback Insert	Valid feedback payload submitted	Record inserted with ID WFQ...	PASS
TC-10	Direct File Execution	frontend/index.html opened via file://	CSS styles and JS logic load cleanly	PASS

CHAPTER 10: CONCLUSION AND FUTURE ENHANCEMENTS

10.1 Conclusion

This project successfully engineered and validated MediSense NLP, a dual-engine healthcare Natural Language Processing framework and full-stack web application tailored for regional Indian hospital ecosystems. By pairing statistical machine learning on 100,000 Warangal hospital reviews with a deterministic clinical report simplifier and Telugu voice synthesizer, the platform concurrently resolves administrative quality-of-care deficits and patient comprehension barriers. The system achieves 93.8% sentiment classification accuracy, cuts diagnostic report readability requirements by over 7 school grades, guarantees 0.00% numerical hallucination, and provides equitable access for non-English speaking populations.

10.2 Future Enhancements

- **Multilingual Dravidian Expansion:** Extend the Dravidian NLG and speech synthesis engine to include Kannada, Tamil, and Malayalam.
- **OCR for Handwritten Prescriptions:** Integrate Optical Character Recognition using Vision Transformers to ingest photographed clinical notes.
- **Offline Edge Android Deployment:** Package the system into an offline Android application using ONNX runtime and offline TTS for rural Primary Health Centers.
- **Federated Healthcare Learning:** Implement federated learning across district hospitals to collaboratively improve grievance classification without sharing patient data.

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